

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Scenario-Based

Course Code:	ITA0532	Course Title:	Computer Vision
CO Assessed:	CO2 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Scenario-Based
Weightage:	10% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)</i>		
Student Name:	Sk. Fawaz Ahamad	Reg. No.:	192425390
Section / Batch:	2024	Date:	10-08-2026

Code of Conduct

I ____Sk. Fawaz Ahamad / 192425390__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____Sk. Fawaz_Ahamad_____

BRICS FOR CALCULATION: ASSESSMENT TOOL 2: Scenario-Based

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	

Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	
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SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

(SIMATS)

Chennai, Tamil Nadu

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B. Tech – Artificial Intelligence & Machine Learning

CO2-AT2

Scenario Based Question

Category	Details
Submitted By	Sk. Fawaz Ahamad
Register No.	192425390
Program	B. Tech Artificial Intelligence & Machine Learning
Year & Semester	III Year
Department	Computer Science and Engineering
Submitted To	Faculty Name: S. christy
Subject	Computer Vision
Academic Year	2026–2027
Date of Submission	10 August 2026

24. Under-Segmentation

Scenario: Important regions are merged incorrectly during image segmentation.

(a) Interpret the scenario and explain under-segmentation

Under-segmentation is a problem in image segmentation where **two or more separate objects or meaningful regions are incorrectly combined into a single segment**. The segmentation algorithm fails to detect the boundary between the regions, resulting in loss of important object information.

For example, consider an image containing **two touching objects** with similar intensity or color. A threshold-based segmentation method may identify both objects as one connected region. Instead of producing two separate regions, the output contains one large merged region.

Example:

Original: Object A Object B

Incorrect segmentation: Object A + Object B $\rightarrow$ One Region

Under-segmentation can reduce the accuracy of subsequent tasks such as **object detection, image analysis, classification, and measurement**.

(b) Identify key issues affecting separation

Several factors can cause under-segmentation:

1. **Similar intensity or colour:** If neighbouring objects have similar pixel values, it becomes difficult for the algorithm to distinguish them.
2. **Weak boundaries:** Poor contrast between adjacent regions can make their boundaries difficult to detect.
3. **Incorrect thresholding:** A poorly selected threshold may classify different objects as the same foreground region.
4. **Touching or overlapping objects:** Objects that are physically connected may appear as one component.
5. **Image noise:** Noise can distort the actual boundaries and affect segmentation accuracy.
6. **Poor image quality:** Low resolution, blur, or uneven illumination can make object separation difficult.

(c) Apply a suitable solution

A suitable solution for this problem is **Watershed Segmentation**, supported by **morphological operations and distance transformation**.

The process can be performed as follows:

1. **Preprocess the image** to reduce noise and improve image quality.
2. Convert the image into **grayscale** if required.
3. Apply **thresholding** to separate the foreground from the background.
4. Use morphological operations such as **opening and closing** to remove noise and improve the segmented regions.
5. Apply a **distance transform** to determine the centre or approximate location of individual objects.
6. Generate markers for the different objects or regions.
7. Apply the **watershed algorithm** to create boundaries between the connected regions.
8. Obtain separate segments for each important object.

This approach is especially useful when objects are **touching or partially overlapping**.

(d) Justify your approach logically

Watershed segmentation is selected because the main problem is that **different regions have been incorrectly joined together**. Simple thresholding may identify the complete connected area but cannot always determine where one object ends and another begins.

The distance transform helps identify the centers of individual objects. These centers are used as markers, and the watershed algorithm then expands the regions until boundaries are formed between them.

Therefore, the method follows a logical sequence:

Noise Removal → Foreground Detection → Object Center Identification → Boundary Formation → Region Separation

This approach provides better separation than using thresholding alone and helps preserve the important structure of individual objects.

(e) Present your explanation clearly

Aspect	Explanation
Problem	Multiple important regions are incorrectly merged
Segmentation error	Under-segmentation
Main causes	Similar intensity, weak boundaries, noise, touching objects and incorrect threshold
Suitable method	Watershed segmentation
Supporting methods	Morphological operations and distance transformation
Process	Preprocessing → Thresholding → Distance Transform → Markers → Watershed
Expected result	Merged objects are separated into individual regions
Benefit	Improves object identification and segmentation accuracy

Conclusion

Under-segmentation occurs when the segmentation algorithm **combines two or more distinct regions into a single region**, causing important information to be lost. It can occur due to similar intensity, weak boundaries, noise, incorrect threshold selection, or touching objects. **Watershed segmentation with morphological preprocessing and distance transformation** provides an effective solution by identifying individual object regions and creating boundaries between them. Hence, the proposed approach improves **region separation, boundary accuracy, and overall image segmentation performance**.